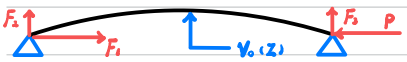
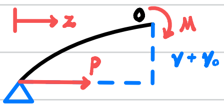
11/26/2023 Euler Buckling Pre-lab

Theory

Imperfections



$F_1 = P \quad F_2 = F_3 = 0 \quad v_0 = v_{c0} \sin(\frac{\pi x}{L})$



$M(O)$, $M = P(v + v_0) = -EI\frac{d^2v}{dz^2}$

$\therefore EI\frac{d^2v}{dz^2} + Pv = -Pv_0 \qquad \therefore \frac{d^2v}{dz^2} + \mu^2 v = -\mu^2 v_0 = -\mu^2 v_{c0}\sin(\frac{\pi z}{L})$ where $\mu^2 = \frac{P}{EI}$

$\because v = A\sin(\mu z) + B\cos(\mu z) + C\sin(\frac{\pi z}{L}) + D\cos(\frac{\pi z}{L})$

$\therefore \frac{d^2v}{dz^2} = -\mu^2[A\sin(\mu z) + B\cos(\mu z)] - (\frac{\pi}{L})^2[C\sin(\frac{\pi z}{L}) + D\cos(\frac{\pi z}{L})]$

substitude back, $(\mu^2 - \frac{\pi^2}{L^2})[C\sin(\frac{\pi z}{L}) + D\cos(\frac{\pi z}{L})] = -\mu^2 v_{c0}\sin(\frac{\pi z}{L})$

$\therefore C = \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2}v_{c0}$, $D = 0$

$\therefore v = A\sin(\mu z) + B\cos(\mu z) + \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2}v_{c0}\sin(\frac{\pi z}{L})$

At $z = 0$, $v = 0$

$B = 0$

At $z = L$, $v = 0$

$A\sin(\mu L) + \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2}v_{c0}\sin(\pi) = A\sin(\mu L) = 0$

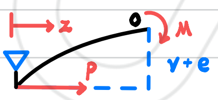
$\therefore \sin(\mu L) = 0$

Let $\mu L = \pi \qquad \therefore (\mu L)^2 = \frac{P}{EI}L^2 = \pi^2 \qquad \therefore P_{crit} = \frac{\pi^2 EI}{L^2}$

At $z = \frac{L}{2}$, $v_c = \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2}v_{c0}\sin(\frac{\pi}{2}) = \frac{\mu^2}{\frac{\pi^2}{L^2} - \mu^2}v_{c0} = \frac{v_{c0}}{\frac{P_{crit}}{P} - 1} = P_{crit}(\frac{v_{crit}}{P}) - v_{c0}$

Eccentricity

$F_1 = P \quad F_2 = F_3 = 0$



$M(O)$, $M = P(v + e) = -EI\frac{d^2v}{dz^2}$

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$\therefore EI\frac{d^2v}{dz^2} + Pv = -Pe \qquad \therefore \frac{d^2v}{dz^2} + \mu^2 v = -\mu^2 e$ where $\mu^2 = \frac{P}{EI}$

$\therefore v = A\sin(\mu z) + B\cos(\mu z) + C$

$\therefore \frac{d^2v}{dz^2} = -\mu^2[A\sin(\mu z) + B\cos(\mu z)]$

substitude back, $\mu^2 C = -\mu^2 e \qquad \therefore C = -e$

$\therefore v = A\sin(\mu z) + B\cos(\mu z) - e$

At $z = 0$, $v = 0$

$\therefore 0 = B - e \qquad \therefore B = e \qquad \therefore A\sin(\mu z) + e[\cos(\mu z) - 1]$

At $z = L$, $v = 0$

$\therefore 0 = A\sin(\mu L) + e[\cos(\mu L) - 1] \qquad \therefore A = \frac{e[1-\cos(\mu L)]}{\sin(\mu L)}$

$\therefore \sin(\mu L) = \cos(\mu L) - 1 = 0 \qquad \therefore \mu L = n\pi$

Let $\mu L = \pi \qquad \therefore (\mu L)^2 = \frac{P}{EI}L^2 = \pi^2 \qquad \therefore P_{crit} = \frac{\pi^2 EI}{L^2}$

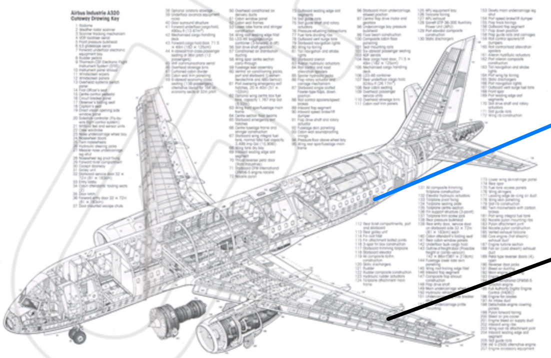
At $z = \frac{L}{2}$, $v_c = A\sin(\frac{\mu L}{2}) + e[\cos(\frac{\mu L}{2}) - 1] = e(\frac{1-\cos(\mu L)}{\sin(\mu L)}\sin(\frac{\mu L}{2}) + \cos(\frac{\mu L}{2}) - 1)$

$$= e(\frac{2\sin^2(\frac{\mu L}{2})}{2\sin(\frac{\mu L}{2})\cos(\frac{\mu L}{2})}\sin(\frac{\mu L}{2}) + \cos(\frac{\mu L}{2}) - 1) = e(\frac{1}{\cos(\frac{\mu L}{2})} - 1) = e[\frac{1}{\cos(\frac{\pi}{2}\sqrt{\frac{P}{P_{crit}}})} - 1]$$

Introduction

Objectives

- caculate P_{crit}
- investigate the effect of changing L, E, A, e and P_{trans}
- compare with theory



slender longerons

stringers

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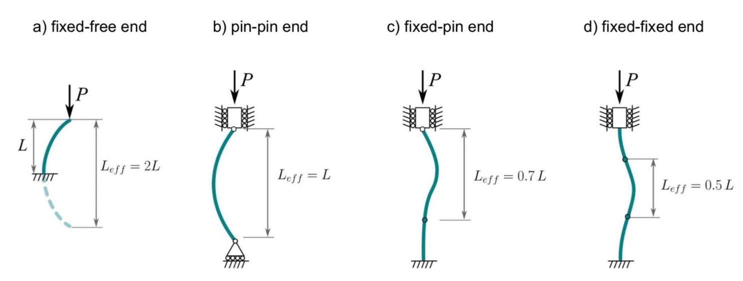
Theory



- $I = \frac{bd^3}{12}$

- $P_{crit} = \frac{\pi^2 EI}{L_{eff}^2}$

Assumptions

- struct is homogeneous E constant
- struct is prismatic A and I constant
- struct is perfectly straight no eccentricity
- compressive load applied to natural axis

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Procedure

Safety

Parameters

Material	Young's Modulus (Nominal)
Steel	207 GN.m^{-2} (207 GPa)
Aluminium	69 GN.m^{-2} (69 GPa)
Brass	105 GN.m^{-2} (105 GPa)

from SM1005 User Guide

Procedure

- Strut 1 - Steel, 20 mm x 3 mm x 750 mm flip
- Strut 2 - Steel, 20 mm x 3 mm x 700 mm +e flip
- Strut 3 - Steel, 20 mm x 3 mm x 650 mm
- Strut 4 - Steel, 20 mm x 3 mm x 625 mm
- Strut 5 - Steel, 20 mm x 3 mm x 600 mm flip
- Strut 6 - Steel, 20 mm x 3 mm x 550 mm +e
- Strut 7 - Brass, 19 mm x 4.8 mm x 750 mm
- Strut 8 - Aluminium, 19 mm x 4.8 mm x 750 mm
- Strut 9 - Steel, 15 mm x 4 mm x 750 mm
- Strut 10 - Steel, 10 mm x 5 mm x 750 mm

- Strut A - Hardwood, 20 mm x 6 mm x 550 mm with steel knife edge inserts.
- Strut B - Plywood, 20 mm x 6 mm x 550 mm with steel knife edge inserts.
- Strut C - Glass Fibre, 20 mm x 5.5 mm x 550 mm with steel knife edge inserts.
- Strut D - Moulded Brass, D shape, 19 mm x 4.5 mm x 550 mm.
- Strut E - Steel Compound, Two steel struts, bolted together. Both 13 mm x 3 mm. One longer (650 mm) with knife edge ends, the other 640 mm.
- Strut F - Aluminium channel, 13 mm x 13 mm and 1.5 mm thick wall x 750 mm with steel end fittings. for fun
- Strut G - Aluminium angle, 13 mm x 13 mm and 1.5 mm thick wall x 750 mm with steel end fittings.
- Strut H - Aluminium angle, 13 mm x 13 mm and 1.5 mm thick wall x 750 mm with steel end fittings.
- Strut I - Steel Rectangular, 13 mm x 6.4 mm x 650 mm.
- Strut J - Steel Round, 6.4 mm diameter x 650 mm.

Data Recording

Deflection

Time	Deflection Indicator		Specimen Properties					DL1 Load Meter	
Time	Deflection	Deflection Position	Material Description	Young's Modulus	Second Moment of Area	Length	End Conditions	Force	Peak Force
(s)	(mm)	(mm)		(GPa)	(mm^4)	(mm)		(N)	(N)
Data Series 1									
0	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
0.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
1	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
1.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
2	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
2.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
3	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
3.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	-1	0
4	0.00	500	Steel	207	0.00	750	Pinned-Pinned	0	0
4.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	5	0
5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	9	0
5.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	9	0
6	0.00	500	Steel	207	0.00	750	Pinned-Pinned	10	0
6.5	0.00	500	Steel	207	0.00	750	Pinned-Pinned	11	0
7	0.07	500	Steel	207	0.00	750	Pinned-Pinned	13	0
7.5	0.10	500	Steel	207	0.00	750	Pinned-Pinned	16	0
8	0.10	500	Steel	207	0.00	750	Pinned-Pinned	19	0
8.5	0.12	500	Steel	207	0.00	750	Pinned-Pinned	20	0
9	0.13	500	Steel	207	0.00	750	Pinned-Pinned	21	0
9.5	0.13	500	Steel	207	0.00	750	Pinned-Pinned	21	0
10	0.18	500	Steel	207	0.00	750	Pinned-Pinned	25	0
10.5	0.21	500	Steel	207	0.00	750	Pinned-Pinned	26	0
11	0.21	500	Steel	207	0.00	750	Pinned-Pinned	28	0
11.5	0.22	500	Steel	207	0.00	750	Pinned-Pinned	28	0
12	0.28	500	Steel	207	0.00	750	Pinned-Pinned	30	0

...... 13 sets of valid data

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Deflection (mm)

Strut 1 Strut 2 Strut 3 Strut 4 Strut 5 Strut 6

Figure 7: Deflection of struts

Strut	1	2	3	4	5	6
Critical Load (N)						

- may not reach buckling load
- load fall for struct 4

abnormal

Strut 1 Strut 2 Strut 5
Strut 1 otherside Strut 2 otherside Strut 5 otherside

Figure 8: Deflection of struts in both side

Strut		1	2	5
Critical Load (N)	Actual			
	Other side			

- P_{crit} not the same

- Could not reach P_{crit}
- stiffness reduced as rate of deflection reduced

Strut 5 Strut 6 with small e Strut 6 with big e

Figure 9: Deflection of struts with eccentric

Measurement

Strut 1 **Steel**

Strut 3 **Steel**

Strut 5 **Steel**

Eccentricity Fitting **Steel**

Strut F - Al Channel **Aluminium**

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Data Analysis

$\frac{1}{L^2}$ vs P_{crit}

Figure 10: $\frac{1}{L^2}$ vs P_{crit}

Strut		1	2	3	4	5	6
Critical Load (N)	Actual						
	Other side						
	Theory						

- E in user guide might inaccurate
- A is different along the struct
- load might too fast

Mode of failure

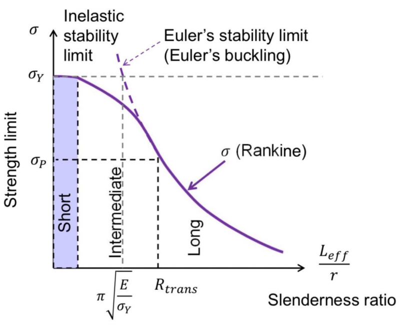

Figure 11: $\frac{L_{eff}}{r}$ vs σ for real struts[2]

Strut	1	2	3	4	5	6
R_{trans}						
Slenderness ratio						
Mode of failure						

- all $\frac{L_{eff}}{r} > R_{trans}$
- struct experience buckling failure

Southwell plot

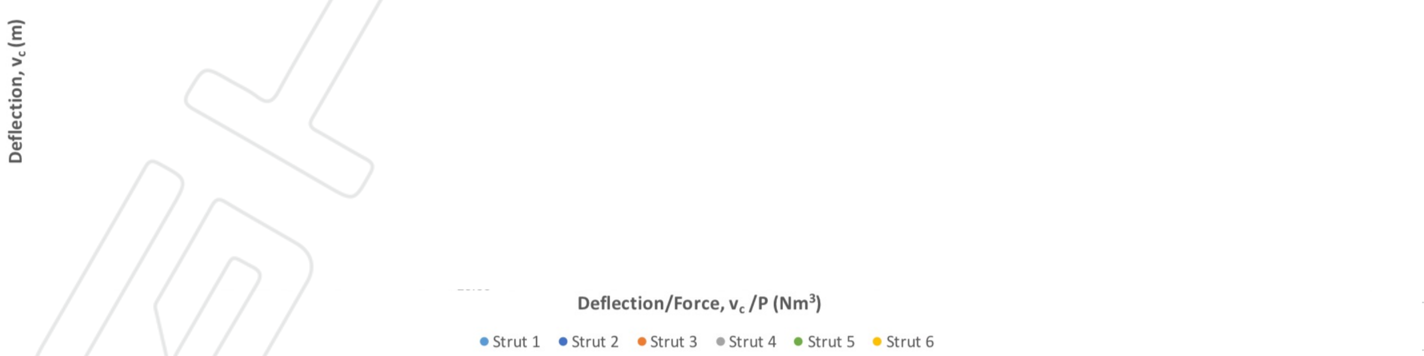

Figure 12: Southwell plot

- no plots cross the origin e exists without adding it

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Buckling Load (N)	Strut 1	Strut 2	Strut 3	Strut 4	Strut 5	Strut 6
Theoretical						
Southwell Plot Method						
Percentage difference (%)						

- $P_{crit}^{southwell} < P_{crit}^{theory}$

+ imperfection

+ alignment error

- do not need to reach P_{crit}

- could obtain imperfections

Figure 13: Southwell plot for Strut 1

Figure 14: Southwell plot for Strut 5

Eccentricity e (mm)	Strut 1		Strut 5	
	Small	Big	Small	Big
Measured from Southwell plot				
Actual known offset				
Difference				

- $e_{measured} > e_{actual}$

⟶ existing e in the struct itself